

EE 4113 – ECE Lab 2
Spring 2026
Lab #02 Report

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Objective:

In this lab exercise, we explore the non-ideal behavior of the wires connecting power and ground to the active elements in a circuit. We find that a “DC” power supply line is NOT a constant DC voltage. We then explore a common way to help “clean-up” the DC power lines by using power supply bypass capacitors (i.e., decoupling capacitors or “bypass caps”).

Equipment List:

Table 1 Equipment List

<i>Description</i>	<i>Manufacturer</i>	<i>Model Number</i>
Solderless Breadboard	R.S.R. Electronics	MB108
Function Generator	Lab-Volt	A8771
Oscilloscope and Probes	Tektronics	DM2510
Power Supply	BK Precision	1762

Parts List:

Table 2 Parts List

<i>Quantity</i>	<i>Part</i>	<i>Value</i>
1	resistor	56 ohm - $\frac{1}{4}$ W
1	resistor	220 ohm - 1 W
2	resistor	1k ohm - $\frac{1}{4}$ W
2	capacitor	0.1 uF
2	electrolyte capacitor	10 uF
1	operational amplifier IC	LF411 or TL071CP

Procedure:

Part A: Circuit Without Bypass Capacitors

- The oscilloscope at the station is prepared by connecting both available probes to separate channels (e.g., Channels 1 and 2), frequency compensating each probe using the oscilloscope’s probe compensation output while ensuring that both probes are configured in the 10× attenuation setting.
- A split power supply is configured to generate ± 12 V, using one supply channel to generate +12 V and the other to generate –12 V, with the metal bridge components used to reference the appropriate terminals to ground, current limits set to an appropriate value (e.g., 0.7 A), and banana-to-hook cables and jumper wires used to establish power and ground connections to the circuit.
- The circuit labeled “Circuit A” is constructed on a breadboard according to the provided schematic and datasheet pin assignments, with a common ground shared between the power supply, function generator, and oscilloscope probes, the ± 12 V rails connected to the appropriate power pins, the input signal applied at V_{in} from the function generator, and the output signal taken from V_{out} , noting that the circuit

functions as a non-inverting amplifier with gain $1 + \frac{R_2}{R_1}$ and an input impedance of approximately $56\ \Omega$.

- The function generator is configured to output a 2 V_{pp}, 100 kHz square wave and connected to the Vin node of the circuit, Channel 1 of the oscilloscope is connected to measure Vin, Channel 2 is connected to measure Vout, and the function generator and power supply are confirmed to be powered on and correctly connected.
- The oscilloscope display is used to verify that an output square wave of approximately 4 V_{pp} is present at Vout, and any wiring or configuration issues are corrected if the expected waveform is not observed.
- Channel 2 of the oscilloscope is then connected to monitor the +12 V power rail, and the channel coupling is set to AC coupling in order to more clearly observe AC noise present on the DC supply line.
- The oscilloscope cursors are used to measure the peak-to-peak magnitude of voltage spikes or oscillations appearing on the +12 V power rail, particularly those synchronized with the rising and falling edges of the square-wave input, and an image of the waveform is saved for inclusion in the lab report.
- After completing the measurements, the function generator output is disabled and the power supply is turned off to conclude Part A.

Part B: Direct Function Generator Measurements with the Oscilloscope vs. Measurements Taken Across a Circuit Load

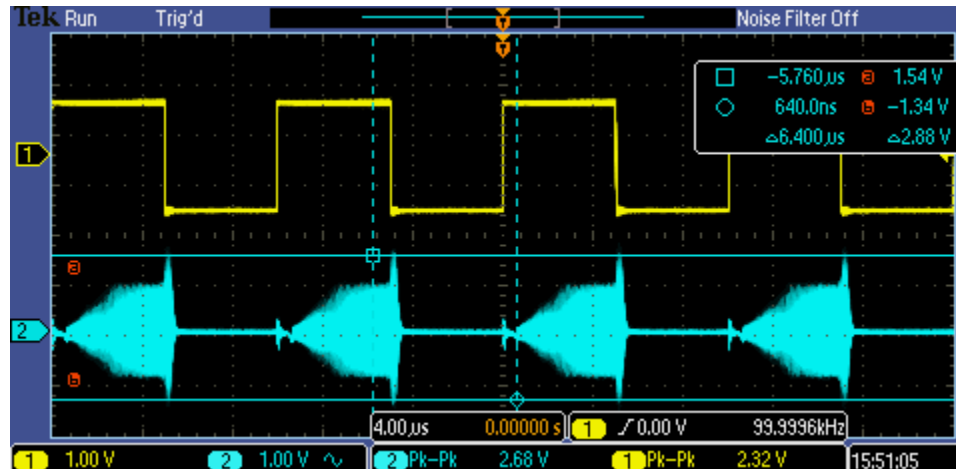
- The circuit constructed in Part A is modified by rebuilding it on a breadboard according to the schematic labeled “Circuit B,” which includes the addition of bypass capacitors C1, C2, C3, and C4 to the power supply rails.
- The smaller decoupling capacitors C1 and C2 are placed physically close to the IC power pins (pins 4 and 7), with their opposite terminals connected as close as possible to the grounded end of the 22 Ω load resistor R_L in order to minimize parasitic inductance along the high-frequency current return path.
- The larger bulk capacitors C3 and C4 are installed with correct polarity, noting that the cathode (negative terminal) of C3 is connected to ground while the cathode of C4 is connected to –12 V, since ground is at a higher potential than –12 V, and while their physical placement is less critical than that of the smaller capacitors, their orientation is essential.
- With the modified circuit in place, the same peak-to-peak measurements performed in Part A are repeated to observe voltage spikes on the +12 V power rail, with the

expectation that the bypass capacitors will attenuate, but not completely eliminate, the high-frequency noise present on the supply line.

- The oscilloscope cursors are used to measure the peak-to-peak magnitude of any remaining voltage spikes or oscillations on the +12 V rail, the vertical scale is increased as much as possible to improve measurement accuracy, and an image of the waveform and measurement is saved for inclusion in the lab report.
- The differences between the voltage spikes measured in Part A and Part B are analyzed in the lab report, with specific attention given to the effectiveness of decoupling capacitors in reducing high-frequency noise on DC power lines and the degree of attenuation achieved through their use.
- After all measurements are completed, the circuit is disassembled, all cables and components are packed away, and the workstation is cleared.

Results:

Part A



Caption:

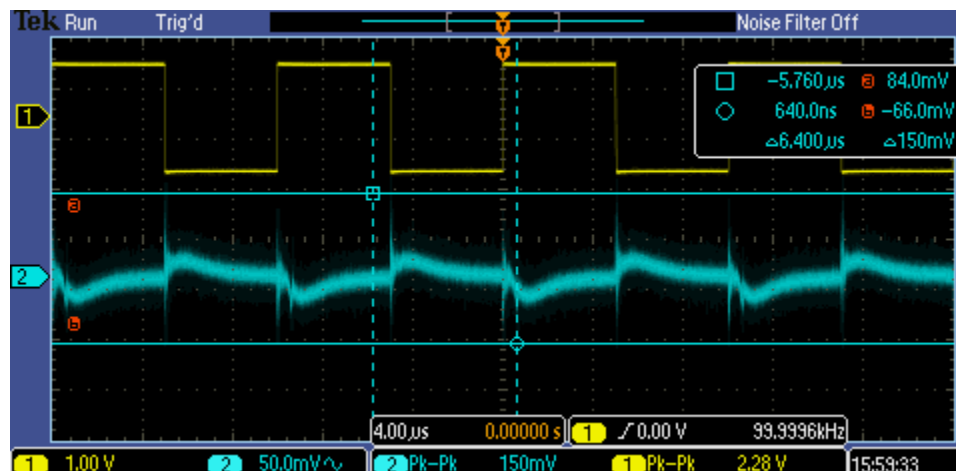
Part A oscilloscope picture pre-capacitor installation.

Description:

This image displays a square wave going into an op-amp and a very noisy signal exiting. The noisy signal is hard to decipher and doesn't give a great sense of values for the voltage at any given point, except for at the end of each spike.

The found peak to peak value is 2.68V.

Part B



Caption:

Part B oscilloscope picture after the capacitors have been installed.

Description:

This image displays the results of putting a square wave signal through an op-amp with capacitors. The output is fairly clean and shows a direct path of increasing and decreasing values. The cursors are also displaying the difference in each peak.

The found peak to peak voltage is 150mV with the cursors displaying the same value.

Conclusion:

The two saved images clearly display the difference that using capacitors in an op-amp can do. Before bypass capacitors were installed, the output had a large amount of noise. Once they were installed for part B the output spikes were simplified significantly and we got a more accurate value of 150mV for the peak to peak voltage. The capacitors reduced the noise and build-up showing the actual value and now attenuated spikes in.